

Cell bodies of the nerves in the autonomic nervous system are located in which region of the spinal cord?

- 7/17/03
- A) Anterior gray horn  
B) Posterior gray horn  
C) Anterior white column  
D) Posterior white column  
E) Lateral gray horn
- 42) The wrong statement about the heart is?
- A) Right ventricle has three papillary muscles.  
B) All valves of the heart have the same shape.  
C) Left ventricle is the thickest of all chambers.  
D) The left atrium has less number of pectinate muscles than the right atrium.  
E) Right atrium forms most of the right border of the heart.
- 43) Which of the following is not artery of the lower limb?
- A) Femoral artery  
B) External iliac artery  
C) Popliteal artery  
D) Fibular artery  
E) Brachial artery

- 44) The vein not located in the abdomen is?
- A) Inferior vena cava  
B) Superior mesenteric vein  
C) Inferior mesenteric vein  
D) Portal vein  
E) Superior vena cava
- 45) The tissues with the highest total glycogen content are:
- A) Muscle and kidneys  
B) Kidneys and liver  
C) Liver and muscle  
D) Brain and Liver  
E) Brain and Muscle

- 46) The cohesion properties of water are caused by:
- A) Ionic bonds  
B) Hydrophobic actions  
C) Hydrogen bonds  
D) Covalent bonds  
E) Salt brigs
- 47) The pH of the extracellular fluid such as blood plasma, of a healthy person is:
- A) Acidic  
B) Slightly acidic  
C) Neutral  
D) Slightly basic  
E) Basic

- 48) A decrease of blood pH from 7.4 to 7.35 would be accompanied by which of the following changes in ion concentration?
- A) A 10-fold increase in hydrogen ion concentration.  
B) A 10-fold increase in hydroxyl ion concentration.  
C) An increase in hydrogen ion concentration by a factor of 4.5/3.5.  
D) A decrease in hydrogen ion concentration by a factor of 3.5/4.5.  
E) A shift in concentration of buffer anions, with no change in hydrogen ion concentration.
- 49) A polymeric unit of starch which has a branched structure is:
- A) Glucose  
B) Amylopectin  
C) Isomaltose  
D) Amylose  
E) Cellobiose

- 50) Two sugars that differ in configuration at only one chiral center are described as:
- A) epimers  
B) diastereomers  
C) hemiketals  
D) enantiomers  
E) anomers

- 51) Which of the following two fatty acids are the most common saturated fatty acids in nature?
- A) Lauric acid and stearic acid  
B) Myristic acid and palmitic acid  
C) Steric acid and arachidic acid  
D) Stearic acid and palmitic acid  
E) Arachidic acid and palmitic acid

in glucose the orientation of the -H and -OH groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determines:

- A) D or L series  
B) Dextro or levorotatory  
C)  $\alpha$  and  $\beta$  anomers  
D) Epimers  
E) Diastereomers

- 53) Infants born prematurely are at risk for respiratory distress syndrome. In such cases, it is common to administer surfactant, the purpose of which is to alter which of the following

- D) They can regulate cellular metabolic activities  
 E) Usually they are part of extracellular matrix
- Heterogeneous nuclear RNA is a precursor of  
 A) Transfer RNA B) Messenger RNA C) Small nuclear RNA  
 D) Ribosomal RNA E) Non-coding RNA
- 68) Which of the following is true about the body fluid regulation  
 A) Antidiuretic hormone increases blood osmolarity  
 B) Diabetes insipidus causes too much water retention  
 C) Administration of hypotonic solution increases fluid in the tissues  
 D) The lower the venous return is the more the ANP  
 E) The more the ANP is the higher the blood pressure
- 69) A 65-year-old man has got infection with severe fever and excessive sweating with more solvent than solute and has become dehydrated. Then which of the following is true about the dehydration and solution administration to this patient.  
 A) The type of dehydration is hypotonic  
 B) The solution administered should be hypertonic  
 C) Colloid containing solution should be administered  
 D) No osmotic gradient created  
 E) Solution hydrating the tissue should be administered
- 70) What is the net filtration pressure through which fluid is derived from the being of the capillary into the tissue of a subject having capillary colloid osmotic pressure of 22 mmHg, capillary hydrostatic pressure of 30 mmHg, tissue colloid osmotic pressure of 10 mmHg and the tissue hydrostatic pressure of -2 mmHg regardless of the filtration coefficient for water and proteins.  
 A) 16 mmHg B) 20 mmHg C) 40 mmHg D) 22 mmHg E) 30 mmHg
- 71) Which of the following is true about the body fluid and electrolytes regulation.  
 A) Atrial natriuretic peptide increases water retention in the body  
 B) Atrial natriuretic peptide is produced when the body fluid volume reduced  
 C) Diabetes insipidus reduces the body fluid volume  
 D) Sodium-pumps activated more when the cellular fluid volume is reduced  
 E) Angiotensin converting enzyme reduces water retention
- A 60-year-old man was admitted into the teaching hospital. Two days following his admission, his extremities became swell and liver cirrhosis was identified by his physician. His serum albumin level was strongly reduced. Which of the following factors contributes the edema more than the other?  
 A) Reduction of the osmotic pressure in the circulation  
 B) Reduction of osmotic pressure in the tissues  
 C) Increase Venous resistance and lymphatic blockage  
 D) Increase the concentration of proteins in the circulation  
 E) Loss of excess proteins through kidney in the form of urine
- 73) Which of the following is true about the negative feedback homeostatic control system?  
 A) The stimuli and the response have the same direction  
 B) Lesley involved in the living organism  
 C) Mostly brings decompensation  
 D) The change made smaller  
 E) Original stimulus is intensified

70/5/03

22  
(30-22)

30-2

(22-10)

22-12  
12

16

(Pc - Pi) - (Pc - Pi)

7/17/23

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$$(30 - 22)$$

$$30 - 2$$

$$(22 - 10)$$

$$22 - 12$$

$$16$$

$$(P_c - P_i) + (a - c_i)$$

$$P_{c1} + P_{c2} + P_{c3}$$

$$P_{c1}$$

- 21) Mitochondria are enclosed by two membranes.  
22) Mitochondria are maternally derived.

23) Concerning the nucleus, which statement is not correct?

- A) Heterochromatin is transcriptionally active.
- B) the outer nuclear membranes may be studded with ribosomes.
- C) Distinction between heterochromatin and euchromatin disappears during cell division.
- D) It detoxifies drugs by oxidation, conjugation and methylation.
- E) It is the largest and essential organelle.

24) Stereocilia

- A) are structurally similar to true cilia
- B) contain 9+2 axonemes ✓
- C) are underlain by basal bodies.
- D) Contain actin filaments in their core ✓
- E) contain nine microtubule triplets at their core.

25) Concerning the cellular morphology, which statement is NOT correct?

- A) Ribosomes are assembled in the nucleolus.
- B) Ribosomes are the sites of protein synthesis.
- C) Centrioles may be located at the base of cilia and flagella.
- D) Centrioles possess low resistance junctions.
- E) The folds on the inner mitochondrial membrane are called cristae.

26) Which of the following statements is NOT correct with regard to cell morphology?

- A) Glycolipids are found in the outer leaflet of a plasmalemma.
- B) Lipid bilayer of plasmalemma is freely permeable to charged ions.
- C) Cholesterol assists in maintaining structural integrity of a cell membrane.
- D) Lysosomes are particularly abundant in cells exhibiting phagocytic activities.
- E) Clathrin-coated vesicles are associated with regulated signal-directed transport of proteins.

27) With regard to the cellular components, which statement is correct?

- A) Lipid bilayer is freely permeable to charged ions.
- B) Microfilaments are more stable than microtubules.
- C) Ribosomes are sites where rRNA is manufactured by the breakdown of  $H_2O_2$  ✓
- D) Barr body is composed of euchromatin during the mitotic phase of a cell cycle.
- E) The nucleolus is a well-defined, membrane-bounded acidophilic nuclear inclusion.

28) Concerning the morphology of a cell, select the correct statement.

- A) Lysosomes self-duplicate in the S phase of the cell cycle. ✓
- B) Stereocilia contain nine microtubule triplets with paired central sheath. ✓
- C) Prophase is the longest and most variable cell-cycle phase in dividing cells. ✓
- D) The intermediate filament vimentin is a marker for connective tissue tumors.
- E) Nucleolus disperses during anaphase but reappears in the metaphase stage of mitosis.

29) Which of the following statements is NOT correct with regard to cell morphology?

- A) Mitochondria proliferate by fission from pre-existing mitochondria.
- B) Ribosomes, as non-membranous organelles, are constructed in the nucleoli.
- C) Hemosiderin is a membrane-bounded vesicle containing oxidative enzymes.
- D) Clathrin-coated vesicles are associated with regulated signal-directed transport of proteins.
- E) Muscle contraction and relaxation is one of the functions of a smooth endoplasmic reticulum.

30) Concerning gametogenesis, which statement is CORRECT?

- A) All oogonia become primary oocytes during oogenesis.

15) Which of the following statements is NOT correct with regard to cell morphology?

- A) rRNA is synthesized by the nucleolus.
- ☒ B) Megakaryocyte is multinucleated cell.
- C) Ion-transporting cells contain abundant basal mitochondria between infoldings of the plasma membrane.
- D)  $G_1$  phase is the longest and most variable cell-cycle phase in continuously dividing cells.
- E) Before DNA replication in the S phase of the cell cycle, chromosomes exist as single-chromatid chromosomes.

16) Concerning structures inside a cell, which statement is correct?

- A) Microtubules are more stable than microfilaments.
- B) SER is the site of assembly of ribosomal subunits.
- C) Synthesis of all proteins appears to be initiated on the free polyribosomes.
- D) Synthesis of all proteins appears to be initiated on the trilaminar unit membranes.
- ☒ E) Glycogen synthesis, storage and breakdown in the liver take place in the mitochondria.

17) With regard to the cellular organelles, which statement is correct?

- A) RER regulates  $H_2O_2$  within cells.
- B) The major function of the mitochondria is the formation of bile salts.
- C) Peroxisomes are membrane-bounded vesicles containing oxidative enzymes.
- D) Nucleolus is a well-defined, membrane-bounded, acidophilic nuclear inclusion.
- ☒ E) Breakdown of fatty acid molecules during beta oxidation takes place in ribosomes.

18) Which of the following pairs of functions is most closely associated with the Golgi complex?

- A) Energy metabolism and glycogen synthesis
- B) Energy metabolism and lipid metabolism
- ☒ C) Glycosylation and sulfation of secretory products
- D) Phagocytosis and receptor recycling
- E) Glycogen synthesis and packaging of secretions.

19) With regard to the nucleolus, select the correct statement.

- A) Nucleolus binds antigens and enzymes to the cell surface.
- B) Nucleolus contains nucleolar organizers where rRNA genes are located.
- ☒ C) Nucleolus aids in cellular attachment of extracellular matrix components.
- D) Nucleolus is generally detectable only when the cell is in early metaphase.
- E) Nucleolus disperses during anaphase but reappears in the metaphase stage of mitosis.

20) With regard to the centrioles, which statement is not correct? Centrioles:

- A) Self-duplicate in the S phase of the cell cycle.
- B) Consist of several disc-shaped cisternae (sacculles) arranged in stacks.
- C) Consist of 9 microtubule triplets linked together in a pinwheel-like arrangement.
- D) Serve as poles during cell division and organize the assembly of the mitotic apparatus.
- ☒ E) Are non-membranous organelles which exist as a pair of cylindrical rods oriented at right angles to one another.

21) Concerning the mitochondria, which statement is false?

- A) Cristae are folds on the inner mitochondrial membrane.
- B) Mitochondrial membranes are continuous with the nuclear envelope.
- C) Mitochondria possess their own genetic apparatus composed of circular DNA.

a network of vessels that withdraws excess tissue fluid from the body's interstitial (intercellular) fluid compartment, filters, and returns it to the bloodstream.

- 1) ☒ A) Cardiovascular system B) Lymphatic system C) Urinary system  
☐ D) Integumentary system E) GI system
- 2) refers to the body position as if the person were standing upright, with the head, gaze (eyes), and toes directed anteriorly (forward). Arms adjacent to the sides with the palms facing anteriorly. Lower limbs close together with the feet parallel and the toes directed anteriorly.  
☐ A) Anatomical position B) Supine position C) Prone position  
☐ D) Liotomy position E) All are correct
- 3) is vertical planes dividing the body into anterior (front) and posterior (back).  
☐ A) Sagittal plane B) Midsagittal plane C) Coronal plane D) Transverse plane  
☐ E) Horizontal
- 4) The word antibrachium refers to  
☒ A) The arm B) The forearm C) The chest D) The nipples E) Arm pit
- 5) The term proximal means  
☒ A) Towards the mid line of the body B) Away from the midline of the body  
☐ C) Toward the trunk of the body D) Away from the trunk of the body E) B and D
- 6) One is type of movement produced at the wrist joint  
☐ A) Adduction B) Medial rotation C) ☒ Supination D) Pronation E) Circumduction
- 7) Movement of the sole of the foot away from the median plane (turning the sole laterally).  
☒ A) Supination B) Pronation C) ☒ Eversion D) Inversion E) Abduction
- 8) is the deep fascia of the thigh, which is the thickest and strongest fascia of the body.  
☒ A) Cole's fascia B) ☒ Fascia lata C) Scarpas fascia D) Campares fascia E) All
- 9) Sharp, slender often pointed projection:  
☐ A) Spine B) ☒ Trochanter C) Process D) Epicondyle E) Crest
- 10) Lesser wing, greater wing, body and pterygoid processes are parts of  
☐ A) Ethmoid bone B) Sphenoid bone C) Temporal bone D) Occipital bone  
☐ E) Zygomatic bone
- 11) Wrist bones of the proximal row arranged from medial to lateral  
☒ A) Scaphoid, Lunate, Triquetrum, Pisiform B) Trapezium, Trapezoid, Capitate, Hamate  
☐ C) Pisiform, Triquetrum, Lunate, Scaphoid D) ☒ Hamate, Capitate, Trapezoid, Trapezium  
☐ E) Capitate, Hamate, Trapezoid, Trapezium
- 12) With regard to the preparation of tissues for microscopic examination, which statement is not correct?  
☐ A) Fixatives are chemical substances used to preserve tissues.  
☐ B) Paraffin is an embedding material.  
☐ C) Eosin stains the mitochondria pink.  
☐ D) Hematoxylin is an acidic dye.  
☒ E) None of the above.
- 13) Which of the following is NOT a membranous organelle?  
☒ A) Lysosomes B) Peroxisomes C) Mitochondria D) Ribosome  
☐ E) Endoplasmic reticulum
- 14) Which one is the characteristic function of a smooth endoplasmic reticulum?  
☐ A) Regulation of  $H_2O_2$  within cells.  
☒ B) ☒ Steroid hormone synthesis.  
☐ C) Gluconeogenesis.  
☐ D) Formation of bile salts.  
☐ E)  $\beta$ -oxidation of long chain fatty acids

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- 53) Infants born prematurely are at risk for respiratory distress syndrome. In such cases, it is common to administer surfactant, the purpose of which is to alter which of the following properties of water at the alveolar interface with air?
- A) Surface tension  
B) Evaporation  
C) Heat of vaporization  
D) Ionization  
E) Dielectric constant

- 74) A young researcher was elevated into high altitude at about 4000 feet from the sea level up a week. However, he has got altitude sickness characterized by fatigue, reduction of appetite, headache. What physiological response made wrong on him
- A) Heart rate increased ✓
  - B) Hemoglobin oxygen carry capacity elevated ✓
  - ~~C) Red blood cell concentration elevated~~
  - D) Respiratory rhythm response increased
  - E) Body fluid volume elevated
- 75) All the following factors contribute to tissue edema except
- A) Increased capillary permeability
  - B) Reduced plasma proteins
  - C) Tissue inflammation
  - ~~D) Increased arterioles resistance~~
  - E) Chronic kidney diseases
- 76) All trigger the release of ADH except
- ~~A) Traumatic burn~~
  - B) Profuse sweating
  - C) Watery diarrhea ✓
  - D) Hypotonic solution infusion
  - E) Sever exercise
- 77) Assume that you have patient with more sodium level in the body fluid than the expected normal value and the patient complains regarding to his/her cardiac and nerve activities. What is wrong on this patient
- A) Reduction of aldosterone action ✓
  - B) Elevation of atrial natriuretic peptide activity ✓
  - C) Elevated renin level in the body
  - D) Reduction of Angiotensin converting enzyme action ✓
  - ~~E) Reduction of Adrenal cortex response~~
- 78) Who have large amount of water content in their body among the following individuals?
- A) Adult males
  - B) Adult females
  - ~~C) Infants~~
  - D) Old males
  - E) Old females
- 79) Select the true concept among the followings
- A) Gain of homeostasis not affected by age and state of subjects
  - B) Loss of more water than solute increase secretion of antidiuretic hormone
  - C) Osmoreceptor activities in the hypothalamus reduced by excess sodium in the blood
  - ~~D) Salivation is one of the ways to gain heat~~
  - E) Increased metabolic activities reduces body temperature ✓
- 80) Which of the following is NOT the function of peroxisomes?
- A) Gluconeogenesis ✓
  - B)  $\beta$ -oxidation of long chain fatty acids. ✓
  - C) Detoxification of ethanol ✓
  - D) Formation of bile salts. ✓
  - ~~E) Destruction of bone matrix.~~